

$$F(A, B, C, D) = \Sigma m(1, 3, 7, 9, 10) + \Sigma d(5, 12)$$

NEW FUNCTION

2. Finding Prime Implicants

Group	Decimal	Binary
0	1	0001 ✓
1	3	0011 ✓
1	9	1001 ✓
1	10	1010 ✖
1	5	0101 ✓
1	12	1100 ✖
2	7	0111 ✓

3. Finding Prime Implicants

Group	Decimal	Binary
0	1, 9	-001 ✖
0	1, 5	0-01 ✓

Group	Decimal	Binary
0	1, 3	00-1 ✓
1	3, 7	0-11 ✓
1	5, 7	01-1 ✓

4. Finding Prime Implicants

Group	Decimal	Binary
0	1, 3, 5, 7	0--1 ✱

The Prime Implicants Are
AB'CD', ABC'D', B'C'D, A'D

6. Prime Implicants Chart

Minterms	1	10	3	7	9	Cost
AB'CD'		X				5
B'C'D	X				X	4
A'D	X		X	X		3

7. Finding Unique Minterms

Minterms	1	10	3	7	9	Cost
$AB'CD'$		X				5
$B'C'D$	X				X	4
$A'D$	X		X	X		3

The Essential Implicants Are
 $AB'CD'$, $B'C'D$, $A'D$

Possible Function Minimizations
 $F = AB'CD' + B'C'D + A'D$